

Limits & Derivatives

ANSWER KEY



Evaluating limits by direct substitution

- 1 Evaluate $\lim_{x \rightarrow 2} (3x^2 - 4x + 1)$.
 $3(4) - 8 + 1 = 12 - 8 + 1 = 5$.
 - 2 Evaluate $\lim_{x \rightarrow -1} (x^3 + 2x - 5)$.
 $(-1)^3 + 2(-1) - 5 = -1 - 2 - 5 = -8$.
 - 3 Evaluate $\lim_{x \rightarrow 0} \frac{5x + 7}{7x + 3}$.
 $\frac{5(0) + 7}{0 + 3} = \frac{7}{3}$.
 - 4 Evaluate $\lim_{x \rightarrow 3} \sqrt{x + 13}$.
 $\sqrt{3 + 13} = \sqrt{16} = 4$.
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Evaluating limits by factoring (the 0/0 form)

- 5 Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{(x - 3)(x + 3)}$.
 $\frac{x^2 - 9}{x - 3} = x + 3 \rightarrow 6$.
 - 6 Evaluate $\lim_{x \rightarrow -2} \frac{x^2 + 5x + 6}{(x + 2)(x + 3)}$.
 $\frac{x^2 + 5x + 6}{x + 2} = x + 3 \rightarrow 1$.
 - 7 Evaluate $\lim_{x \rightarrow 4} \frac{x^2 - 16}{(x - 4)(x + 4)}$.
 $\frac{x^2 - 16}{x - 4} = x + 4 \rightarrow 8$.
 - 8 Evaluate $\lim_{x \rightarrow 1} \frac{x^2 - 1}{(x - 1)(x + 1)}$.
 $\frac{x^2 - 1}{(x - 1)(x - 2)} = \frac{x + 1}{x - 2} \rightarrow \frac{2}{-1} = -2$.
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Limits at infinity

- 9 Evaluate $\lim_{x \rightarrow \infty} \frac{3x^2 + 2x}{x^2 - 5}$.
The degrees match, so the limit is the ratio of leading coefficients: 3.
 - 10 Evaluate $\lim_{x \rightarrow \infty} \frac{2x + 7}{x^2 + 1}$.
The denominator has higher degree, so the limit is 0.
 - 11 Evaluate $\lim_{x \rightarrow \infty} \frac{5x^3 - 2}{x^3 + 4x}$.
The degrees match, so the limit is the ratio of leading coefficients: 5.
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One-sided limits

- 12 A function is defined by $f(x) = x + 1$ for $x < 2$, and $f(x) = 2x - 1$ for $x \geq 2$. Find $\lim_{x \rightarrow 2^-} f(x)$.
Using the first rule: $\lim_{x \rightarrow 2^-} f(x) = 2 + 1 = 3$.
- 13 Using the same piecewise function, find $\lim_{x \rightarrow 2^+} f(x)$.
Using the second rule: $\lim_{x \rightarrow 2^+} f(x) = 2(2) - 1 = 3$.
- 14 Using the same piecewise function, does $\lim_{x \rightarrow 2} f(x)$ exist? State its value.
Yes — both one-sided limits equal 3, so $\lim_{x \rightarrow 2} f(x) = 3$.
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The power rule

- 15 Differentiate $f(x) = x^5$.
 $f'(x) = 5x^4$.
- 16 Differentiate $f(x) = 3x^4 - 2x^2 + 7$.
 $f'(x) = 12x^3 - 4x$.
- 17 Differentiate $f(x) = \sqrt{x}$.
 $f(x) = x^{1/2}$, so $f'(x) = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$.
- 18 Differentiate $f(x) = \frac{1}{\sqrt{x}}$.
 $f(x) = x^{-1/2}$, so $f'(x) = -\frac{1}{2}x^{-3/2} = -\frac{1}{2x\sqrt{x}}$.
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The product rule

- 19 Differentiate $f(x) = x^2(3x + 1)$ using the product rule.
 $f'(x) = (2x)(3x + 1) + x^2(3) = 6x^2 + 2x + 3x^2 = 9x^2 + 2x$.
- 20 Differentiate $f(x) = (x + 2)(x - 5)$ using the product rule.
 $f'(x) = (1)(x - 5) + (x + 2)(1) = x - 5 + x + 2 = 2x - 3$.
- 21 Differentiate $f(x) = (2x - 1)(x^2 + 3)$ using the product rule.
 $f'(x) = (2)(x^2 + 3) + (2x - 1)(2x) = 2x^2 + 6 + 4x^2 - 2x = 6x^2 - 2x + 6$.
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The chain rule

- 22 Differentiate $f(x) = (3x + 1)^4$.
 $f'(x) = 4(3x + 1)^3(3) = 12(3x + 1)^3$.

- 23 Differentiate $f(x) = (x^2 - 5)^3$.
 $f'(x) = 3(x^2 - 5)^2(2x) = 6x(x^2 - 5)^2$.
- 24 Differentiate $f(x) = \sqrt{4x + 9}$.
 $f(x) = (4x + 9)^{1/2}$, so $f'(x) = \frac{1}{2}(4x + 9)^{-1/2}(4) = \frac{2}{\sqrt{4x + 9}}$.
- 25 Differentiate $f(x) = (5 - 2x)^6$.
 $f'(x) = 6(5 - 2x)^5(-2) = -12(5 - 2x)^5$.
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Derivatives of trigonometric and exponential functions

- 26 Differentiate $f(x) = \sin x + \cos x$.
 $f'(x) = \cos x - \sin x$.
- 27 Differentiate $f(x) = 3\sin x$.
 $f'(x) = 3\cos x$.
- 28 Differentiate $f(x) = e^x + x^2$.
 $f'(x) = e^x + 2x$.
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Tangent lines

- 29 Find the slope of the tangent line to $f(x) = x^2$ at $x = 3$.
 $f'(x) = 2x$, so $f'(3) = 6$.
- 30 Find the equation of the tangent line to $f(x) = x^2$ at $x = 2$.
 $f(2) = 4$, $f'(2) = 4$. $y - 4 = 4(x - 2) \Rightarrow y = 4x - 4$.
- 31 The graph of $f(x) = x^2 - 2x$ is shown. Find the equation of the tangent line at $x = 3$ (the marked point).
 $f(3) = 9 - 6 = 3$. $f'(x) = 2x - 2$, so $f'(3) = 4$. $y - 3 = 4(x - 3) \Rightarrow y = 4x - 9$.
- 32 Find the x -coordinate(s) where the tangent to $f(x) = x^3 - 3x$ is horizontal.
 $f'(x) = 3x^2 - 3 = 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$.
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Increasing, decreasing, and the second derivative

- 33 For $f(x) = x^2 - 4x + 1$, find $f'(x)$ and determine the interval where f is increasing.
 $f'(x) = 2x - 4$. Increasing when $f'(x) > 0$, i.e. $x > 2$.
- 34 For $f(x) = -x^2 + 6x$, determine the interval where f is decreasing.
 $f'(x) = -2x + 6$. Decreasing when $f'(x) < 0$, i.e. $-2x + 6 < 0 \Rightarrow x > 3$.

- 35 Find the second derivative of $f(x) = x^4 - 2x^2$.
 $f'(x) = 4x^3 - 4x$, so $f''(x) = 12x^2 - 4$.
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Word problems: rates of change

- 36 This question has two parts. The height of a ball thrown upward is given by $h(t) = -5t^2 + 20t + 2$ (in meters, t in seconds).
- a Find the velocity function $h'(t)$. $h'(t) = -10t + 20$.
- b Find the velocity at $t = 1$ second, and state whether the ball is rising or falling.
 $h'(1) = -10(1) + 20 = 10$ m/s, which is positive, so the ball is rising.
- 37 This question has two parts. A company's cost to produce x items is $C(x) = 0.01x^2 + 5x + 200$ (in dollars).
- a Find the marginal cost function $C'(x)$. $C'(x) = 0.02x + 5$.
- b Find the marginal cost at $x = 100$ items, and interpret its meaning.
 $C'(100) = 0.02(100) + 5 = 2 + 5 = 7$ dollars — approximately the cost of producing one more item beyond the 100th.